

M9415B VXT PXIe Vector Transceiver with Analog BBIQ

330 MHz to 20 GHz

Introduction

The Keysight M9415B VXT PXIe vector transceiver is a compact multi-functional solution designed for both RF and baseband I/Q (BBIQ) signal analysis. It integrates a 1.2 GHz bandwidth vector RF transceiver and baseband I/Q transceiver into a 4-slot PXIe module, enables the end-to-end design validation and troubleshooting from RF to baseband I/Q thoroughly and cost-effectively.

This configuration guide contains information to help you configure your M9415B VXT vector transceiver with analog BBIQ to meet your requirements.

Ordering optional capabilities at the time of purchasing the base product provides the lowest overall cost.



Included in Base Product

Standard options and accessories come with the VXT base model at no additional charge and do not need to be ordered. They include:

- Option F08: Frequency range, 330 MHz to 8.5 GHz
- Option B4X: 400 MHz modulation and analysis bandwidth
- Option M02: Memory, 256 MSa
- Option BBA: Analog baseband I/Q
- N9060EM0E I/Q analyzer
- Cable, SMB female to SMP female, 300 mm
- PXI Modular Product Start-up Reference

Hardware

Refer to the following steps to configure the M9415B VXT:

A. Select Options for VXT Vector Transceiver

Step 1. Choose Your Module		
M9415B-001	VXT PXIe Vector Transceiver with Analog BBIQ	Licensed as RCV, SRC, BBA
Step 2. Choose Frequency Range		
M9415B-F08	Frequency range, 330 MHz to 8.5 GHz	Included in base configuration
M9415B-F13	Frequency range, 330 MHz to 13.5 GHz	Optional
M9415B-F20	Frequency range, 330 MHz to 20 GHz	Optional
Step 3. Choose Modulation and Analysis Bandwidth		
M9415B-B4X	Bandwidth, 400 MHz	Included in base configuration
M9415B-B8X	Bandwidth, 800 MHz	Optional
M9415B-B12	Bandwidth, 1.2 GHz	Optional
Step 4. Choose Memory Size		
M9415B-M02	Memory, 256 MSa	Included in base configuration
M9415B-M05	Memory, 512 MSa	Optional
Step 5. Choose Additional Capabilities		
M9415B-1EA	High output power	Optional
M9415B-RPC	5G NR real-time phase compensation	Optional
M9415B-MDM	External modulation input and demodulation output	Optional

B. Add M9300A PXIe Frequency Reference

Step 1. Add an M9300A PXIe Frequency Reference (Occupies 1 Slot)

One Frequency Reference Required per System to Meet Data Sheet Specifications

M9300A	PXIe frequency reference	Five 100 MHz outputs; One 10 MHz output Internal 10 MHz OCXO timebase output
M9300A-S01 ¹		

C. Select Controller

Step 1. Choose Embedded Controller²

M9035A-M16 PXIe embedded controller, Intel i3-8100H quad-core processor, 3.0 GHz, 4-thread, 16 GB RAM

M9038A-M32 High-performance embedded controller, Intel i7-9850HE 6-core processor, 2.7 GHz, 12-thread, 32 GB RAM, with two Thunderbolt 3.0 ports

Select M9038A for the best performance if you have memory intensive applications, multiple applications running in parallel, or if a lot of data is sent to the PC from the PXIe chassis. Features removable SSD drive for security and multiple connectors from front for connection to second chassis



Step 2. Upgrade from Standard Memory Size (Optional)

M9035A-M32	Memory upgrade to 32 GB RAM
M9038A-M64	Memory upgrade to 64 GB RAM

Step 3. Select Operating System

N9035A-W11	Windows 11 IoT Enterprise LTSC (64-bit)
M9038A-W11	Windows 11 IoT Enterprise LTSC (64-bit)

1. No export license required
2. The M9010A 10-slot or M9019A 18-slot chassis includes empty space to the left of the 1st functional slot. The embedded controller occupies that empty space and the 1st functional slot.

To Use Your Desktop PC as a Controller ^{1, 2}

M9048B	PCIe host adapter: Single port (x8), Gen 3	
M9049A	PCIe host adapter: Dual port (x16), Gen 3	
Y1202A	PCIe cable	
M9021A ³	PCIe cable interface: Gen 2, x8	
M9022A	PXIe system module: Single Port (x8), Gen 3	
M9023A	PXIe system module: Dual Port (x16), Gen 3	
M9024A	PCIe cable interface: Gen 2, x8	

PC Requirements for Controlling M9415B VXT ⁴

Functions	Description
Operating system	Windows 11 (64 bit)
Processor speed	1.86 GHz dual core minimum
Available memory	8 GB minimum 16 GB is required for LTE or 5G NR measurement
Available disk space on Drive C	16 GB minimum 40 GB recommended for multiple applications

1. For list of qualified external controllers, please see Test Computer List Technical Note literature no. 5990-7632EN.
2. For more detailed chassis configuration information including multi-chassis, see Interface Modules and Adapters for PXIe and AXIe Systems literature no. 5992-0377EN.
3. The M9021A can only be used with the Keysight M9018B.
4. For list of qualified external controllers, please see Test Computer List Technical Note literature no. 5990-7632EN.

D. Select A Chassis and Accessories

Step 1. Select a Chassis

M9010A	10-slot PXle chassis, Gen 3	
M9019A	18-slot PXle chassis, Gen 3	
M9046A	High power, 18-slot PXle chassis, Gen 3	

Step 2. Choose Enough Slot Block Kits and EMC Filler Panels to Fill Every Open Slot. Recommended to Achieve Data Sheet Specifications

Y1212A	Slot blocker kit: 5 slots	
Y1213A	PXI EMC filler panel kit: 5 slots	
	Non-EMC filler panels are included with the M9010A, M9018B, or M9019A PXle chassis.	

Step 3. Choose a Rack Mount Kit ¹ (Optional)

Y1271A	Rack mount kit for M9010A and Y1217A rail kit	
Y1215C	Rack mount kit for M9019A 18-slot PXle chassis	
Y1216B	Rack mount kit for M9019A 18-slot PXle chassis	

Step 4. Choose an Air Inlet Kit (Optional). Recommend for Rack Mounted Systems with Less than 1U Space Below Chassis

Y1214B	Air inlet kit: M9019A 18-slot chassis	
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1. For more information on the rack mount kit, see the chassis data sheet, literature number 5992-1481EN.

Step 5. Select Accessories

Y1823A	Calibration kit for Baseband I/Q analyzer, with 8 pieces of SMB-SMB cables and USB cable	Required ¹
Y1824A	Cable Assembly-Coaxial A07/A36 50-Ohm, SMB-Female to SMP-Female, 300 mm	Optional
Y1824B	Cable Assembly-Coaxial A07/A36 50-Ohm, SMB-Female to SMP-Female, 500 mm	Optional
Y1824C	Cable Assembly-Coaxial A36/A36 50-Ohm, SMP-Female to SMP-Female, 200 mm	Optional
Y1824D	Cable Assembly-Coaxial A07/A07 50-Ohm, SMB-Female to SMB-Female, 1000 mm	Optional
Y1824E	Cable Assembly-Coaxial A13/A36 50-Ohm SMA-Male to SMP-Female, 1000 mm	Optional
Y1824F	Cable Assembly-Coaxial 50-Ohm, SMB-Plug to SMB-Plug, 350 mm	Optional
Y1824G	Cable Assembly-Coaxial 50-Ohm, 3.5 mm-Plug to 3.5 mm-Plug, 36-inch, 26.5GHz-Max	Optional
Y1824H	Cable Assembly-Coaxial A07/A13 50-Ohm, SMB-Female to SMA-Male, 760 mm	Optional
Y1814A	Cable Assembly-Coaxial 50-Ohm, SMA to SMA, 1220 mm	Optional
Y1825A	Adapter-coaxial straight Jack-SMB to Jack SMA, 50-ohm, 4 GHz-Max	Optional
Y1826A	USB3.1 3m to Type C, 0.5 meter	Optional

1. One unit of Y1823A is required per test set.

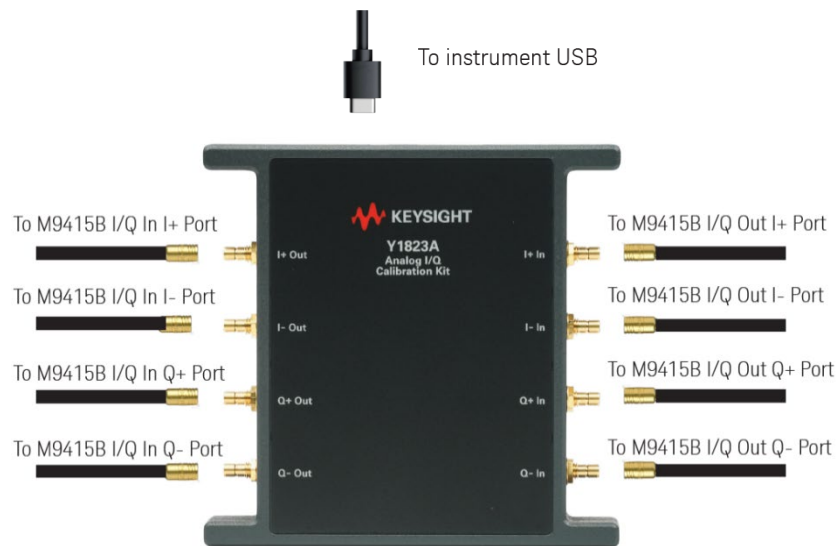


Figure 1. Y1823A Calibration kit for Analog baseband I/Q

Software

Select measurement applications or software and license type for VXT PXIe vector transceiver.

The VXT comes standard with the following software:

- Keysight IO Libraries Suite including Connection Expert
- Drivers for use with MATLAB, Visual Studio (including VB Net, C#, C/C++), Keysight VEE
- Sample waveforms and programming examples
- N9060EM0E I/Q analyzer measurement application

Step 1. Add X-Series Measurement Applications (Optional)

Measurement applications that start with the prefix "N", or "E" are measurement only applications, and require Signal Studio license or waveform pack licenses for waveform playback.

Measurement applications that start with the prefix "Y" combines measurement applications with unlimited waveform playback capability

Keysight offers 4 license types for the measurement applications. Each of the following license types are offered as Perpetual or Subscription. Visit www.keysight.com/find/X-Series_apps for more information.

- Node-locked: Allows you to use the license on one specified instrument or computer.
- Transportable: Allows you to move the license from one instrument or computer to another using Keysight's online tool.
- USB portable: Allows you to move the license from one instrument or computer to another with a certified USB dongle.
- Floating: Allows you to access the license on networked instruments or computers from a server, one at a time.

N9054EM0E E9054EM0E	Vector modulation analysis digital demodulation measurement application
N9054EM1E E9054EM1E	Vector modulation analysis custom OFDM measurement application
N9055EM0E E9055EM0E	Power amplifier measurement application
N9077EM0E E9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah measurement application
N9077EM1E E9077EM1E	WLAN 802.11/ac/ax measurement Application
N9077EM2E E9077EM2E	WLAN 802.11be and 802.11bn measurement application
N9080EM0E E9080EM0E	LTE and LTE-Advanced FDD measurement application
N9080EM3E E9080EM3E	NB-IoT and eMTC FDD measurement application
N9080EM4E E9080EM4E	LTE-V2X measurement application
N9081EM0E E9081EM0E	Bluetooth® measurement application
N9082EM0E E9082EM0E	LTE and LTE-Advanced TDD Measurement application
N9085EM0E E9085EM0E	5G NR Measurement application
N9073EM0E E9073EM0E	W-CDMA/HSPA+ measurement application
N9071EM0E E9071EM0E	GSM/EDGE/Evo measurement application

Step 2. Add Signal Studio Software ¹ (Optional)

N7606EMBC	Bluetooth, waveform playback
N7608EMBC	Custom modulation, waveform playback
N7617EMBC	WLAN 802.11, waveform playback
N7624EMBC	LTE / LTE-Advanced FDD, waveform playback
N7625EMBC	LTE / LTE-Advanced TDD, waveform playback
N7626EMBC	Signal Studio for V2X, waveform playback
N7600EMBC	Signal Studio for W-CDMA/HSPA+, waveform playback
N7602EMBC	Signal Studio for GSM/EDGE/Evo, waveform playback
N7631EMBC	5G NR, waveform playback
N7650B	5/50 pack license for supported formats

1. For more information, see Signal Studio brochure, literature number 5989-6448EN or Signal Studio configuration assistant at http://rfmw.em.keysight.com/wireless/helpfiles/all-in-one_config_asst/ssconfig.html.

Besides ordering on X-series measurement applications and Signal Studio, alternatively, you may also order at Y-applications, that are bundles of signal analysis and Signal Studio licenses.

Step 3. Buy Software Bundles (Optional)

Y9077EM0E	WLAN 802.11a/b/g/j/p/n Waveform and Measurement Application
Y9077EM1E	WLAN 802.11ac/ax Waveform and Measurement Application
Y9077EM2E	WLAN 802.11be/bn Waveform and Measurement Application
Y9080EM0E	LTE and LTE-Advanced FDD Waveform and Measurement Applications
Y9080EM3E	NB-IoT and eMTC FDD Waveform and Measurement Application
Y9080EM4E	LTE V2X Waveform and Measurement Application
Y9081EM0E	Bluetooth TM Waveform and Measurement Application
Y9082EM0E	LTE and LTE-Advanced TDD Waveform and Measurement Applications
Y9085EM0E	5G NR Waveform and Measurement Application
Y9071EM0E	GSM/EDGE/Evo Waveform and Measurement Application
Y9073EM0E	W-CDMA/HSPA+ Waveform and Measurement Application

Step 4. Add More Software (Optional)

N6171A	MATLAB software - Create arbitrary waveforms, customize measurement and data analysis routines, create your own instruments applications and test systems, automate measurements, signal generation, and report generation. Visit www.keysight.com/find/n6171a for more information.
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Calibration

Model Number	Description	
M9415B-UK6	Commercial calibration certificate with test data	Only available for ordering along with base instrument purchase
M9300A-UK6	Commercial calibration certificate with test data	
R-50C-011-3-D	Keysight calibration – Return to Keysight – 3 years	
R-50C-011-5-D	Keysight calibration – Return to Keysight – 5 years	

Global Warranty

Keysight provides the peace of mind that today's high-tech industry requires. Your investment is protected by Keysight's global reach in more than 100 countries (either directly or through distributors). The warranty gives you convenient standard coverage for the country in which the product is used, eliminating the need to ship equipment back to the country of purchase.

Keysight's warranty service provides:

- All parts and labor necessary to return your investment to full specified performance
- Recalibration for products supplied originally with a calibration certificate
- Return shipment

Keysight Support Services

Accelerate your learning curve, enhance your test uptime, and confidently guarantee your instrument accuracy with Keysight Support Services. Keysight Services are here to support your test needs with expert technical support, instrument repair and calibration, training, alternative acquisition program options, and more.

A KeysightCare agreement provides dedicated, proactive support through a single point of contact for an extensive group of instruments, software, and solutions to ensure optimal uptime, with fast response times and resolution. Explore the services that are right for you.

Keysight Services

Offering	Benefits
KeysightCare 	KeysightCare provides elevated support for Keysight instruments and software, with access to technical support experts who respond within a specified time and ensure committed repair and calibration turnaround times (TAT). KeysightCare offers multiple service agreement tiers, including KeysightCare Assured, Enhanced, and Application Software Support. See the KeysightCare data sheet for details.
KeysightCare Assured	KeysightCare Assured provides a commitment to respond to your engineer's technical needs quickly. When unexpected repairs are necessary, you can count on a committed repair service turnaround time to get you back up and running.
KeysightCare Enhanced	KeysightCare Enhanced includes all the benefits of KeysightCare Assured plus Keysight's accurate and reliable Calibration Services , accelerated and committed TAT, and technical response.
Keysight Support Portal & Knowledge Center	All KeysightCare tiers include access to the Keysight Support Portal, where you can manage support and service resources related to your assets, such as service requests and status, or browse the Knowledge Center.
Education Services	Build confidence and gain new skills to make accurate measurements, with flexible Education Services developed by Keysight experts. Including Start-up Assistance.
Alternative Acquisition Options	
KeysightAccess	Reduce budget challenges with a lease-based subscription service that offers low monthly payments, enabling you to get the instruments, software, and technical support you want for your test needs.

Recommended Services

Maximize your instrument uptime and confidently make accurate measurements by securing technical support, repair, and calibration services with committed response and turnaround times. High-performance instruments include 1 year of KeysightCare Assured or KeysightCare Warranty Plus. Obtain multi-year KeysightCare upfront to eliminate the need for lengthy and tedious paperwork and yearly requests for maintenance budget. Plus, you benefit from secured service for 2, 3, or 5 years.

Service	Function
KeysightCare Enhanced*	Includes Tech Support, Warranty and Calibration
R-55B-001-1	KeysightCare Enhanced – Upgrade 1 year
R-55B-001-2	KeysightCare Enhanced – Extend to 2 years
R-55B-001-3	KeysightCare Enhanced – Extend to 3 years (Recommended)
R-55B-001-5	KeysightCare Enhanced – Extend to 5 years (Recommended)
KeysightCare Assured*	Includes Tech Support and Warranty
R-55A-001-2	KeysightCare Assured – Extend to 2 years
R-55A-001-3	KeysightCare Assured – Extend to 3 years
R-55A-001-5	KeysightCare Assured – Extend to 5 years
Start-Up Assistance	
PS-S40-01	Included – instrument fundamentals and operations starter
PS-S40-04	Recommended – instrument fundamentals and operations starter
PS-S40-02	Optional, technology & measurement science standard learning

* Limited availability might apply. Please review the [service definition tool](#) for model number availability and the [datasheet](#) for country availability. Coverage might be limited to KeysightCare Warranty Plus (R-55F-001). If KeysightCare Enhanced is available. R-55B-001-2/3/5 must be ordered with R-55B-001-1.

Using a Non-Keysight Chassis

The M9415B VXT PXIe vector transceiver can be successfully installed in a non-Keysight PXI chassis. Please refer to the following guidelines:

- Ensure that the chassis has enough consecutive PXIe or PXI-H slots to accommodate the M9415B.
- Ensure that the chassis and controller support peer-to-peer PXI Express I/O switch topology.
- Ensure that controller selected is compatible with chassis.

Please contact your Keysight representative for more detailed information. For technical assistance with non-Keysight equipment, please refer to the equipment manufacturer's website.

Related Literature

For more detailed product and specification information refer to the following literature and web pages:

Publication Title	Publication Number
M9415B VXT Vector Transceiver with Analog BBIQ – Data Sheet	3126-1101EN
M9010A PXIe 10-slot chassis - Data Sheet	5992-1707EN
M9018B and M9019A PXIe 18-slot chassis - Data Sheet	5992-1481EN
M9046A High Power, PXIe 18-slot chassis – Data Sheet	3122-1479EN
M9038A PXIe Embedded Controller - Data Sheet	3122-1717EN
X-Series Measurement Applications - Brochure	5989-8019EN
Simplify Signal Creation with Signal Studio Software - Brochure	5989-6448EN

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